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Uses:

(i) Freons are a used as refrigerants in refrigerators and air conditioners.

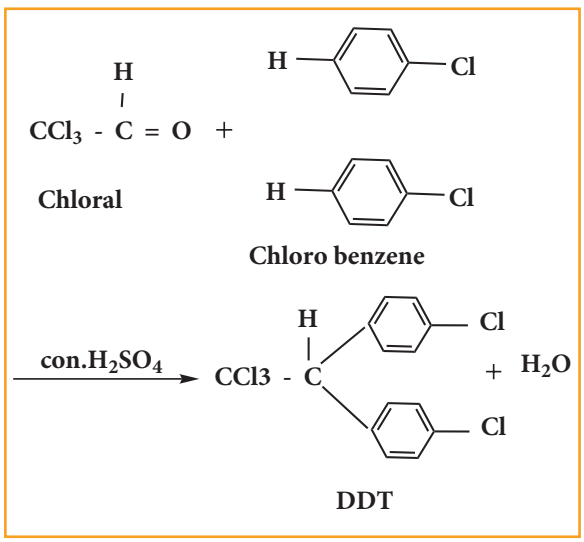
(ii) It is used as a propellant for aerosols and foams

(iii) It is used as propellant for foams to spray out deodorants, shaving creams, and insecticides.

14.6.5 DDT (p,p'-dichloro diphenyl tri-chloro ethane)

DDT, the first chlorinated organic pesticide was prepared in 1873, and in 1939 Paul Muller discovered the effectiveness of DDT as an insecticide. He was awarded Noble prize in medicine and physiology in 1948 for this discovery.

DDT can be prepared by heating a mixture of chlorobenzene with chloral (Trichloro acetaldehyde) in the presence of con.H_2SO_4.



Evaluate Yourself

8) What is the IUPAC name of the insecticide DDT? Why is their use banned in most of the countries?

Uses:

i) DDT is used to control certain insects which carries diseases like malaria and yellow fever

ii) It is used in farms to control some agricultural pests

iii) It is used in building construction as pest control

iv) It is used to kill various insects like housefly and mosquitoes due to its high and specific toxicity.

SUMMARY

- The compounds obtained by the substitution of hydrogen atom of alkanes by halogen atom are called haloalkane, while the compounds obtained by the substitution of hydrogen atoms of arenes by halogen atom are called haloarenes.
- Modern classification of halo compounds is based on the halogen with carbon possessing sp^3 hybridisation. In these compounds the electronegativity of halogen is more than that of carbon, hence $C\delta+ - X\delta-$ bond becomes polar.
- Haloalkane
- Haloalkanes are prepared from alkanes, alkenes or alcohols. The boiling points of haloalkane are higher than that of corresponding hydrocarbons.